

AIM:

13 Performing linear regression analysis using lm() (R).

INPUT:-

```
library(readr)
```

```
shades <- read_csv("shades.csv")
```

```
shades_30 <- shades[1:30, ]
```

```
model <- lm(L ~ H, data = shades_30)
```

```
summary(model)
```

```
plot(shades_30$H, shades_30$L,  
      xlab = "Hue (H)",  
      ylab = "Lightness (L)",  
      main = "Linear Regression: Hue vs Lightness ",  
      pch = 19,  
      col = "blue")
```

```
abline(model, col = "red", lwd = 2)
```

OUTPUT:-

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```
Console Terminal Background Jobs
R v4.1.2 . ~/
-- Column specification -----
Delimiter: ","
chr (5): brand, brand_short, product, product_short, hex
dbl (5): H, S, V, L, group

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
> shades_30 <- shades[1:30, ]
> model <- lm(L ~ H, data = shades_30)
> summary(model)

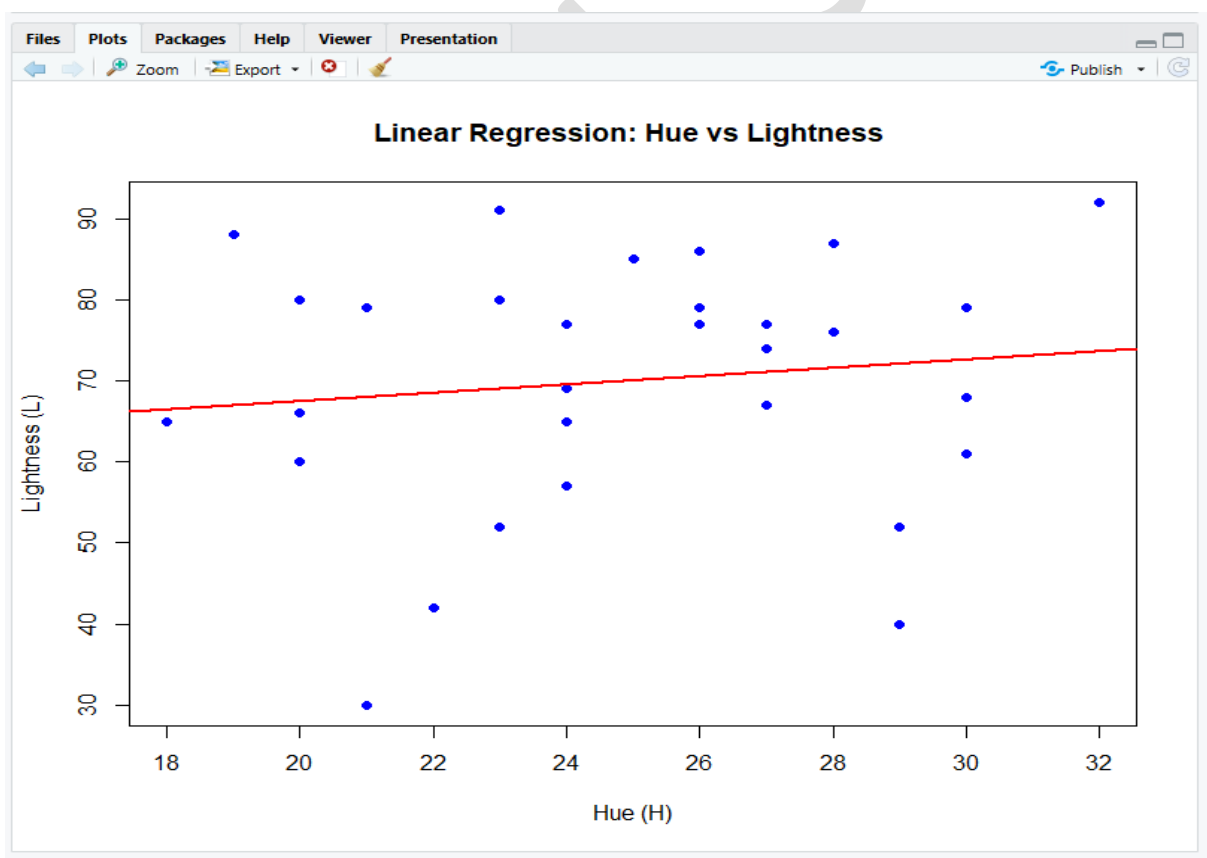
Call:
lm(formula = L ~ H, data = shades_30)

Residuals:
    Min       1Q   Median       3Q      Max
-38.029  -6.807   3.602  10.961  21.934

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  57.1462    19.6166   2.913  0.00696 **
H              0.5183     0.7804   0.664  0.51206
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 15.71 on 28 degrees of freedom
Multiple R-squared:  0.01551, Adjusted R-squared:  -0.01965
F-statistic: 0.441 on 1 and 28 DF, p-value: 0.5121

>
> plot(shades_30$H, shades_30$L,
+       xlab = "Hue (H)",
+       ylab = "Lightness (L)",
+       main = "Linear Regression: Hue vs Lightness ",
+       pch = 19,
+       col = "blue")
> abline(model, col = "red", lwd = 2)
> |
```



14 Performing logistic regression using glm() (R).

INPUT:-

```
library(ggplot2)
```

```
library(dplyr)
```

```
wine <- read.csv("D:/S079_VIBHUTI/ADV PYTHON FOR DATA SCIENCE/winequality-red.csv")
```

```
names(wine) <- tolower(names(wine))
```

```
names(wine) <- gsub("_", ".", names(wine))
```

```
wine$high_quality <- ifelse(wine$quality >= 7, 1, 0)
```

```
wine$high_quality <- as.factor(wine$high_quality)
```

```
model <- glm(
```

```
  high_quality ~ alcohol,
```

```
  data = wine,
```

```
  family = binomial
```

```
)
```

```
summary(model)
```

```
ggplot(wine, aes(x = alcohol, y = as.numeric(high_quality) - 1)) +
```

```
  stat_smooth(
```

```
    method = "glm",
```

```
    method.args = list(family = "binomial"),
```

```
    se = FALSE,
```

```
    color = "red",
```

```
    size = 1.5
```

```
) +
```

```
labs(
```

```
  x = "Alcohol",
```

y = "Probability of High Quality Wine",

title = "Logistic Regression: Alcohol vs High Quality Wine"

) +

theme_minimal() +

theme(

plot.title = element_text(hjust = 0.5, face = "bold", size = 14),

axis.title = element_text(face = "bold")

)

OUTPUT:-

```
> library(ggplot2)
> library(dplyr)
>
> wine <- read.csv("D:/S079_VIBHUTI/ADV PYTHON FOR DATA SCIENCE/winequality-red.csv")
>
> names(wine) <- tolower(names(wine))
> names(wine) <- gsub("_", ".", names(wine))
>
> wine$high_quality <- ifelse(wine$quality >= 7, 1, 0)
> wine$high_quality <- as.factor(wine$high_quality)
>
> model <- glm(
+   high_quality ~ alcohol,
+   data = wine,
+   family = binomial
+ )
>
> summary(model)
```

Call:

glm(formula = high_quality ~ alcohol, family = binomial, data = wine)

Deviance Residuals:

Min	1Q	Median	3Q	Max
-2.2379	-0.5054	-0.3211	-0.2482	2.6822

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-13.23466	0.84022	-15.75	<2e-16 ***
alcohol	1.05057	0.07486	14.03	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance:	1269.9	on 1598	degrees of freedom
Residual deviance:	1027.9	on 1597	degrees of freedom
AIC:	1031.9		

Number of Fisher Scoring iterations: 5

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```
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R 4.1.2 ~/

glm(formula = high_quality ~ alcohol, family = binomial, data = wine)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-2.2379  -0.5054  -0.3211  -0.2482   2.6822

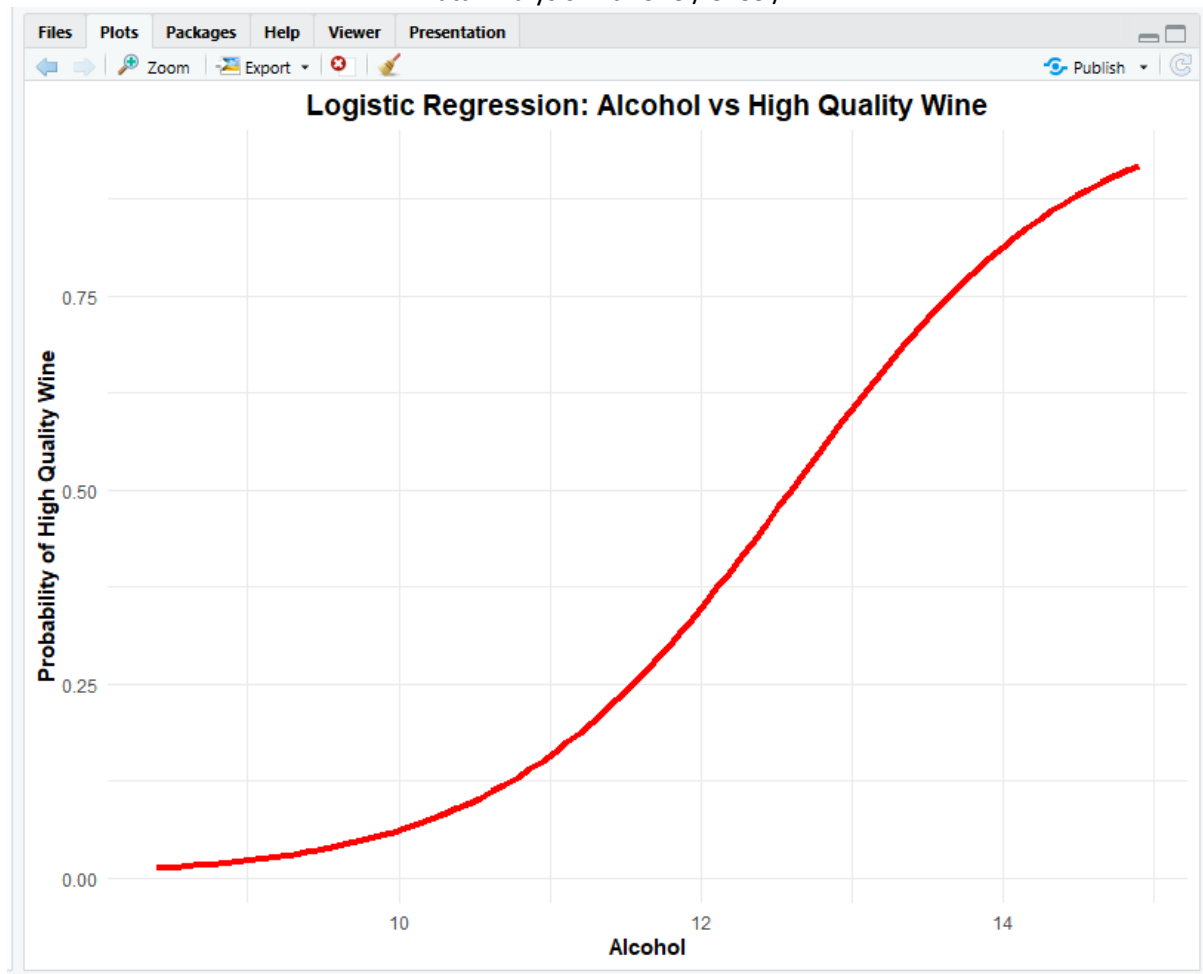
Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) -13.23466    0.84022  -15.75  <2e-16 ***
alcohol       1.05057    0.07486   14.03  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 1269.9  on 1598  degrees of freedom
Residual deviance: 1027.9  on 1597  degrees of freedom
AIC: 1031.9

Number of Fisher Scoring iterations: 5

>
> ggplot(wine, aes(x = alcohol, y = as.numeric(high_quality) - 1)) +
+   stat_smooth(
+     method = "glm",
+     method.args = list(family = "binomial"),
+     se = FALSE,
+     color = "red",
+     size = 1.5
+   ) +
+   labs(
+     x = "Alcohol",
+     y = "Probability of High Quality wine",
+     title = "Logistic Regression: Alcohol vs High Quality wine"
+   ) +
+   theme_minimal() +
+   theme(
+     plot.title = element_text(hjust = 0.5, face = "bold", size = 14),
+     axis.title = element_text(face = "bold")
+   )
+   `geom_smooth()` using formula = 'y ~ x'
> |
```



15 Exporting results into external files (Excel, CSV, PDF) using write.csv() and writexl (R).

INPUT:-

```
library(ggplot2)
```

```
wine <- read.csv("D:/S079_VIBHUTI/ADV PYTHON FOR DATA SCIENCE/winequality-red.csv")
```

```
names(wine) <- tolower(names(wine))
```

```
names(wine) <- gsub("_", ".", names(wine))
```

```
wine$high_quality <- ifelse(wine$quality >= 7, 1, 0)
```

```
wine$high_quality <- as.factor(wine$high_quality)
```

```
model <- glm(
```

```
  high_quality ~ alcohol,
```

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```
data = wine,  
family = binomial  
)
```

```
coef_table <- as.data.frame(summary(model)$coefficients)
```

```
write.csv(coef_table, "D:/S079_VIBHUTI/ADV PYTHON FOR DATA  
SCIENCE/logistic_model_coefficients.csv", row.names = TRUE)
```

```
write.csv(wine, "D:/S079_VIBHUTI/ADV PYTHON FOR DATA SCIENCE/wine_processed.csv",  
row.names = FALSE)
```

```
wine$predicted_prob <- predict(model, type = "response")
```

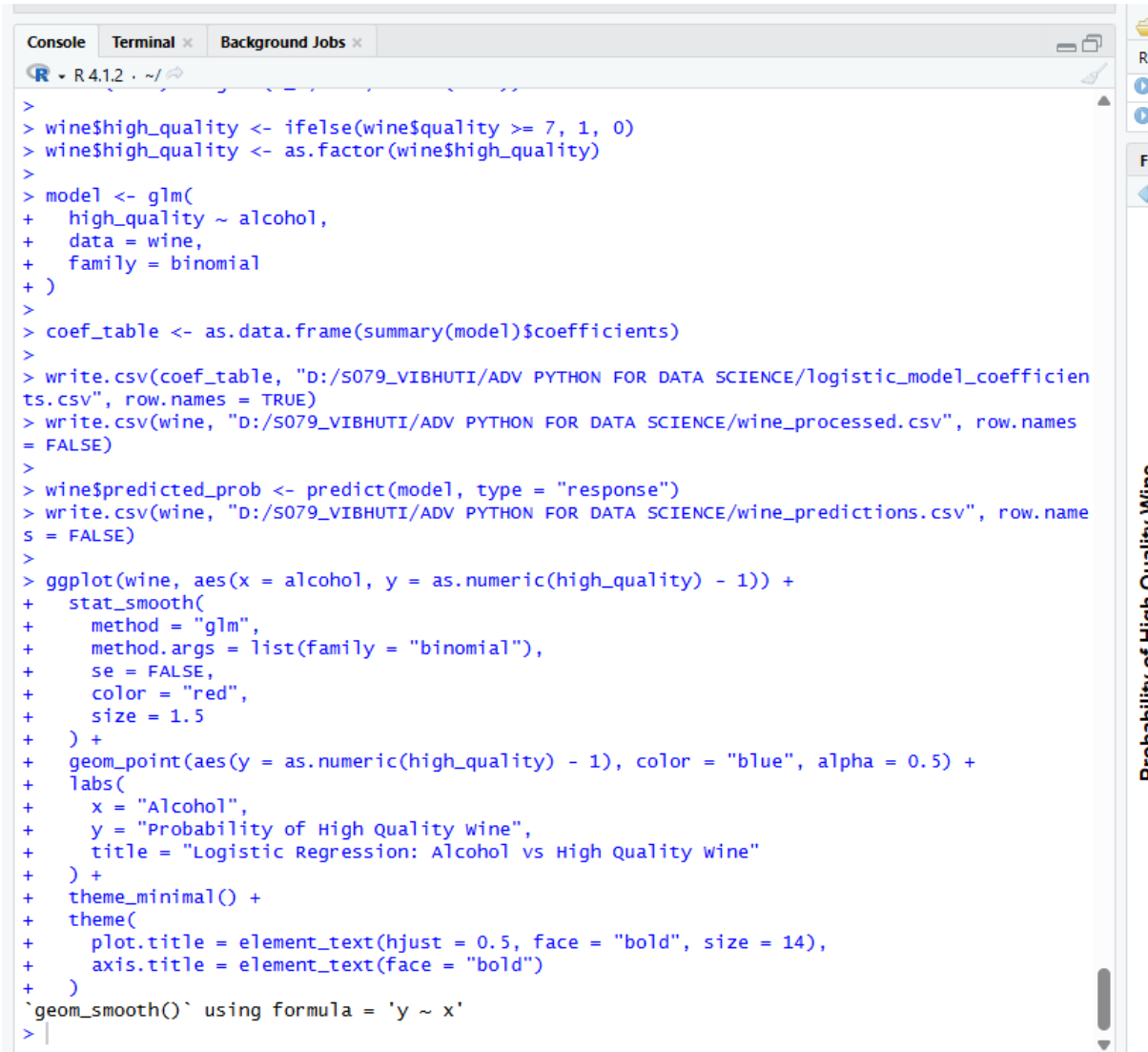
```
write.csv(wine, "D:/S079_VIBHUTI/ADV PYTHON FOR DATA SCIENCE/wine_predictions.csv",  
row.names = FALSE)
```

```
ggplot(wine, aes(x = alcohol, y = as.numeric(high_quality) - 1)) +  
  stat_smooth(  
    method = "glm",  
    method.args = list(family = "binomial"),  
    se = FALSE,  
    color = "red",  
    size = 1.5  
  ) +  
  geom_point(aes(y = as.numeric(high_quality) - 1), color = "blue", alpha = 0.5) +  
  labs(  
    x = "Alcohol",  
    y = "Probability of High Quality Wine",  
    title = "Logistic Regression: Alcohol vs High Quality Wine"  
  ) +  
  theme_minimal() +  
  theme(  
    plot.title = element_text(hjust = 0.5, face = "bold", size = 14),
```

```
axis.title = element_text(face = "bold")
```

```
)
```

OUTPUT:



```
>
> wine$high_quality <- ifelse(wine$quality >= 7, 1, 0)
> wine$high_quality <- as.factor(wine$high_quality)
>
> model <- glm(
+   high_quality ~ alcohol,
+   data = wine,
+   family = binomial
+ )
>
> coef_table <- as.data.frame(summary(model)$coefficients)
>
> write.csv(coef_table, "D:/S079_VIBHUTI/ADV PYTHON FOR DATA SCIENCE/logistic_model_coefficients.csv", row.names = TRUE)
> write.csv(wine, "D:/S079_VIBHUTI/ADV PYTHON FOR DATA SCIENCE/wine_processed.csv", row.names = FALSE)
>
> wine$predicted_prob <- predict(model, type = "response")
> write.csv(wine, "D:/S079_VIBHUTI/ADV PYTHON FOR DATA SCIENCE/wine_predictions.csv", row.names = FALSE)
>
> ggplot(wine, aes(x = alcohol, y = as.numeric(high_quality) - 1)) +
+   stat_smooth(
+     method = "glm",
+     method.args = list(family = "binomial"),
+     se = FALSE,
+     color = "red",
+     size = 1.5
+   ) +
+   geom_point(aes(y = as.numeric(high_quality) - 1), color = "blue", alpha = 0.5) +
+   labs(
+     x = "Alcohol",
+     y = "Probability of High Quality wine",
+     title = "Logistic Regression: Alcohol vs High Quality wine"
+   ) +
+   theme_minimal() +
+   theme(
+     plot.title = element_text(hjust = 0.5, face = "bold", size = 14),
+     axis.title = element_text(face = "bold")
+   )
`geom_smooth()` using formula = 'y ~ x'
>
```


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> library(cupify)
> wine <- read.csv("D:/S079_VIBHUTI/ADV PYTHON FOR DATA SCIENCE/winequality-red.csv")
> names(wine) <- tolower(names(wine))
> names(wine) <- gsub("_", ".", names(wine))
> wine$high_quality <- ifelse(wine$quality >= 7, 1, 0)
> wine$high_quality <- as.factor(wine$high_quality)
> model <- glm(
+   high_quality ~ alcohol,
+   data = wine,
+   family = binomial
+ )
> summary(model)

Call:
glm(formula = high_quality ~ alcohol, family = binomial, data = wine)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-2.2379  -0.5054  -0.3211  -0.2482   2.6822

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) -13.23466    0.84022  -15.75  <2e-16 ***
alcohol       1.05057    0.07486   14.03  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

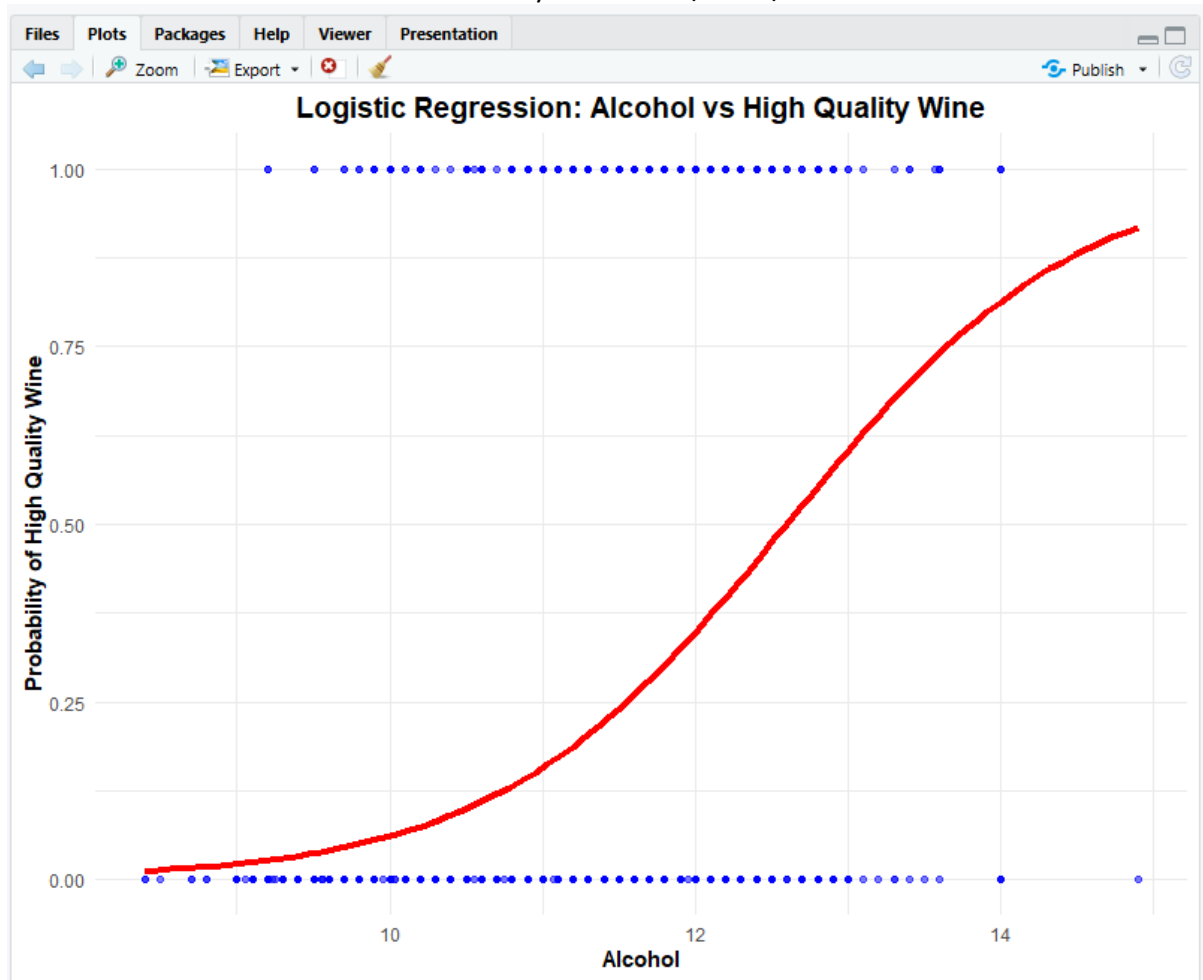
(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 1269.9  on 1598  degrees of freedom
Residual deviance: 1027.9  on 1597  degrees of freedom
AIC: 1031.9

Number of Fisher Scoring iterations: 5

>
> ggplot(wine, aes(x = alcohol, y = as.numeric(high_quality) - 1)) +
+   stat_smooth(
```

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The image shows an Excel spreadsheet with the following data:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-13.2347	0.840221	-15.7514	6.72E-56
alcohol	1.05057	0.074863	14.03326	9.76E-45